International Journal of Sports Physiology and Performance, (Ahead of Print)
https://doi.org/10.1123/ijspp.2021-0343
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First Published Online: Feb. 24, 2022

Human Kinetics
ORIGINAL INVESTIGATION

Postmatch Recovery Practices Carried Out in Professional Football: A Survey of 56 Portuguese Professional Football Teams

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Background: In football (ie, soccer), the effectiveness of recovery practices is critical to maintain high levels of performance, prevent residual fatigue, and reduce injury risk. However, the recovery methods that professional football teams put in place up to 72 hours postmatch are widely unknown. ***Purpose:*** This study aimed to characterize the postmatch recovery practices in Portuguese football teams in different postmatch periods and match locations. ***Methods:*** Portuguese football teams (total, N = 56; male: first league [n = 17], second league [n = 16], under-23 league [n = 12]; female: first league, n = 11) participated in the study. The practitioners in charge of recovery strategies in each team filled out the online questionnaire in the middle of the 2019–20 season. The questions focused on the type of recovery methods to be used at different times after home and away matches. ***Results:*** After home matches, stretching, electrostimulation, active recovery, and massage were used with higher frequency ($P < .017$) in later postmatch periods (ie, 12–24, and 24–72 h) compared with immediately postmatch. After away matches, several differences were found ($P < .017$) for the stretching, electrostimulation, active recovery, cold-water immersion, massage, nutrition, and sleep between postmatch periods. Regarding match location, stretching ($r = .19$), active recovery ($r = .39$), cold-water immersion ($r = .46$), and massage ($r = .29$) showed a higher frequency of use immediately after home matches. Conversely, the use of compression garments ($r = .27$) was higher immediately after away matches. ***Conclusions:*** It was concluded that in professional football, recovery methods are not applied uniformly along postmatch periods and differ depending on the match location.

Keywords: questionnaire, recovery assessment, fatigue, elite football, soccer

The demands of a football (ie, soccer) match have a deleterious impact on physiological homeostasis.[1] As a consequence, players' physical capacities are known to be impaired, at least, up to 72 hours postmatch, with signs and symptoms of muscle damage peaking at 48 hours postmatch.[1–3] This effect might have a greater impact on elite than nonelite players, due to more congested competitive schedules and greater physiological demands of match play.[1,4] Therefore, optimal recovery postmatch is fundamental to avoid long-term fatigue and adverse consequences such as under-recovery, nonfunctional overreaching, or overtraining syndrome.[2,5] Previous systematic reviews related to recovery topics in elite adult football players were conducted to provide knowledge and guidelines to sport scientists and coaches.[2,5–11] However, to our knowledge, just one systematic review has evaluated the methodological quality of the analyzed studies, and has developed graded recommendations for practice.[8] Moreover, information about the choice, the perception of effectiveness, the frequency of use, and the periodization of recovery methods in practical contexts is still scarce.

In the last decade, several studies have examined recovery methods in teams sports using questionnaires based on athletes' perception[12–15] and on professionals in charge of athletes' postmatch recovery.[2,3,16,17] Among these studies, only 3 were conducted exclusively with elite football teams[2,16,17] and only one examined the use of recovery methods after match.[17] Overall, studies performed in elite football have shown that a great importance was attached to recovery methods to mitigate fatigue and enhance recovery.[2,16,17] In addition, food/fluid replacement, supplementation, sleep, active recovery, and cold water immersion were the most used recovery methods.[2,16,17] However, since the recovery period can last (at least) up to 72 hours,[1–3] it is important to examine whether the recovery methods are differently used within this period. To our knowledge, only Field et al[17] have partially examined this issue, and reported that, within an in-season 7-day microcycle, recovery strategies are used most frequently on match day and the day after the match.

Moreover, it is known that the choice of recovery methods could also be influenced, for instance, by the match location.[7,10,17] Away matches have been reported to require higher team external loading (eg, total distance covered) and induce greater mental and physical fatigue.[7,18] In addition, traveling effects on away matches may favor higher fatigue and can potentially make it difficult to implement recovery methods.[7,10,17] However, no previous studies have examined whether the use of postmatch recovery practices differ depending on the match location.

Therefore, the present study aimed to characterize postmatch recovery practices in Portuguese football teams in different postmatch periods (ie, immediately after, 12–24 h, and 24–72 h after matches), and match locations (ie, home and away). In addition, we analyzed whether associations exist between the order of importance attached to each recovery method and their degree of usage. We hypothesized that: (1) recovery methods usage would be greater immediately after and 12 to 24 hours postmatch than 24 to 72 hours postmatch, (2) away matches would promote greater

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use of recovery methods in postmatch periods, and (3) an association would be observed between the order of importance attached and the degree of recovery methods usage.

Methods

Subjects

All Portuguese top tier male (first league [n = 18]; second league [n = 18], and under-23 league [n = 16] and female [first league, n = 12]) football teams (n = 64) were invited to participate in the study. Invitations were addressed to the practitioners in charge of athletes' postmatch recovery, and only one respondent and response for each team was allowed and considered for analysis. The eligibility criteria of participants were: (1) to be a practitioner with an academic background and experience in sports coaching, sports science, physical therapy, or sports medicine and (2) to be responsible for decision making about the type and dose of postmatch recovery practices in the respective club. Only participants who agreed to participate in the study and fulfilled all mandatory questions were considered. Regarding the recommendations to define a representative sample size in a known population (ie, 64 practitioners), a cutoff of 55 respondents was considered with a 5% margin of error and 95% confidence interval.[19] All participants provided written informed consent to participate in the study.

Study Design

A follow-up study was conducted in order to fulfill the purposes of the current investigation. A previously designed and validated questionnaire, containing 19 questions, divided into 6 sections, was applied to assess the recovery practices within 72 hours postmatch in Portuguese professional football clubs.[20] Briefly, the questionnaire contained sections with questions for informed consent, personal information, to examine the postmatch importance, and to characterize the recovery method type and the frequency at different periods after home and away matches. Most questions were considered mandatory (ie, required to answer) with exceptions for C2, D2 (ie, ranking questions), and "other" options. The applied questionnaire in Portuguese language and the translation in English (unvalidated version) can be found in the Supplementary Data (File 1) (available online). The study was approved ethics committee of the Faculty of Human Kinetics, University of Lisbon (approval number: 10/2019), and the procedures were conducted according to the principles expressed in the Declaration of Helsinki.

Procedures

The aforementioned questionnaire,[20] written in Portuguese language, was inserted in an online survey platform (LimeSurvey Open-Source platform, version 3.17.9; LimeSurvey GmbH, Hamburg, Germany), and the content ensured the anonymity of the participants.

The electronic version of the file can be accessed in Supplementary Data (File 2) (available online). Official invitations to participate in the study were performed by the researchers' institutional email contacts between January and February 2020. A continuous follow-up was performed by email during this period to remind the participants to fill in the questionnaire, and to ensure high participation. The questionnaire was rendered inactive at the end of February 2020, and responses were downloaded. All data were exported from LimeSurvey platform to Excel files and organized for subsequent statistical analysis. For ranking questions (ie, recovery methods, overall importance, and frequency of usage), 1 point was summed for each rank position, where 9 points was attributed to the first rank place, and 1 point was given to the ninth place. Then, the total number of points for a given recovery method was summed and considered for analysis. For the remaining questions examining the recovery methods' importance and frequency usage (ie, 5-point Likert scale), 1 point was summed for each rank position. This means that 1 point was attributed to the "not important or never used" classification, and 5 points was attributed to the "extremely important or always used" classification (ie, the least and highest ranking positions, respectively).

Statistical Analyses

All data were analyzed using IBM SPSS Statistics (version 26; IBM Corporation, New York, NY). Given the absence of normal distribution of all variables analyzed by Kolmogorov–Smirnov test, all statistical analyses were computed using nonparametric tests. The frequency of usage difference between postmatch periods (ie, immediately after, 12–24 h, and 24–72 h) of each method was determined using the nonparametric Friedman test. Wilcoxon signed-rank test was performed as a post hoc test, with Bonferroni correction conducted, to avoid type I errors, resulting in a significance level set at $P < .017$. To compare the recovery methods frequency of usage between match locations (ie, home and away matches) at a given time period, a nonparametric Wilcoxon signed-rank test was performed for a $P < .05$. As recommended, the magnitude of the standardized effect size between periods and between locations were calculated for each recovery method using a correlation coefficient (r) and interpreted as small (.10), medium (.30), and large (.50).[21,22] The Spearman correlation test was applied to determine the correlation coefficient (ρ) between the order of importance and the use of recovery methods. This correlation analysis provides additional and complementary information in which one could understand whether the most important methods are those more often used, and the other way around.

Results

A total of 56 teams (88%) participated in the study, which represents a 4.7% margin of error, with a 95% confidence level. The questionnaire responders were fitness coaches (n = 27; 48%), physiotherapists (n = 17; 30%), team physicians (n = 8; 15%), head coaches (n = 3; 5%), and nurses (n = 1; 2%). Half of the sample (n = 28) reported having more than 5 years of professional experience with high-performance team sports.

A similar order of importance and frequency of usage for recovery methods were observed, except for the 2 first positions (Table 1). Specifically, sleep was ranked first and nutrition second, while the opposite was observed for frequency of usage. A positively large correlation was observed concerning the importance order of recovery methods and the degree of use for each recovery method ($r = 1.00$; $P < .01$).

All teams assumed the usage of recovery methods as fundamental for the recovery process, and 63% (n = 35) reported adopting different recovery strategies depending on the time of the football match (ie, morning, afternoon, and evening). Also, 68% (n = 38) of respondents reported the need to change the recovery methods in away matches when the traveling duration was higher than 4 hours. The overall degree of recovery methods usage showed differences ($P < .05$) depending on the different postmatch periods and the match location (Figure 1). Some of the other

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Table 1 Ranking of Importance Given and Frequency of Use of Recovery Methods, According to the Summed Points Obtained in Ranking Positions Ordered by Participants

	Importance given		Frequency of use	
Ranking	Method	Points	Method	Points
1	Sleep	507	Nutrition	443
2	Nutrition	479	Sleep	432
3	Active recovery	421	Active recovery	420
4	Cold-water immersion	361	Cold-water immersion	355
5	Massage	263	Massage	254
6	Stretching	217	Stretching	243
7	Compression garments	190	Compression garments	134
8	Electrostimulation	154	Electrostimulation	133
9	Other	57	Other	49

recovery methods (not mentioned in the survey) were suggested by the study participants: foam roller (n = 3), cognitive therapy (n = 1), contrast water therapy (n = 3), and hot water therapy (n = 2).

Use of Recovery Methods After Home Matches

After home matches, a higher frequency of usage ($P < .017$) was observed for active recovery ($r = .46$), electrostimulation ($r = .33$), massage ($r = .30$), and stretching ($r = .36$) in the period 12 to 24 hours than immediately postmatch. In the period 24 to 72 hours compared with immediately after matches, higher frequency of usage was noted ($P < .017$) for active recovery ($r = .54$), electrostimulation ($r = .43$), massage ($r = .45$), and stretching ($r = .50$).

Use of Recovery Methods After Away Matches

After away matches, a higher frequency of usage ($P < .017$) for active recovery ($r = .52$), electrostimulation ($r = .32$), massage ($r = .45$), stretching ($r = .35$), and others ($r = .23$) was observed in the period 12 to 24 hours than immediately after the matches. In contrast, the frequency of nutrition ($r = .26$) was slightly higher ($P < .017$) immediately after matches than in the period 12 to 24 hours after. In the period 24 to 72 hours after compared with immediately after away matches, higher frequency of usage ($P < .017$) was reported for active recovery ($r = .58$), cold water immersion ($r = .30$), electrostimulation ($r = .37$), massage ($r = .53$), sleep ($r = .24$), stretching ($r = .51$), and others ($r = .29$). In respect to the period 24 to 72 hours, compared with the period 12 to 24 hours postmatch, higher frequency of usage ($P < .017$) was reported for active recovery ($r = .39$), cold water immersion ($r = .23$), electrostimulation ($r = .23$), massage ($r = .35$), and stretching ($r = .38$).

Recovery Methods Differ According to Match Location

Regarding the differences of recovery methods usage between match locations, higher frequency of use immediately after home matches was reported ($P < .05$) for active recovery ($r = .39$), cold water immersion ($r = .46$), massage ($r = .29$), and stretching ($r = .19$). In contrast, the use frequency of compression garments ($r = .27$) was slightly higher ($P < .05$) immediately after away matches. In the period 12 to 24 hours after matches, higher frequency of use was observed ($P < .05$) in home matches for active recovery ($r = .24$) and stretching ($r = .22$). Finally, in the period 24 to 72 hours after matches, stretching ($r = .19$, $P < .05$) was reported to be used more after home matches.

Discussion

This study aimed to characterize the recovery practices used postmatch in Portuguese male and female football teams. Fifty-six football teams participated in the study, which constitutes 88% of all Portuguese top tier football teams. Although the participants reported the great importance and use of several recovery methods to mitigate postmatch fatigue, it was clear that nutrition and sleep were the most important and most used methods. It was also observed that recovery methods varied considerably between postmatch periods and match location. In addition, a large positive association between the order of the importance and the use of each recovery method was observed. This indicates that the importance given by practitioners to recovery methods (which is influenced by their knowledge and experience of the effectiveness of recovery methods) is translated into the frequency of usage (which implementation may be influenced by, for instance, logistical issues).

Use of Recovery Methods in Different Postmatch Periods

Although research is limited on this topic, the need for periodization of recovery methods in football has been suggested.[10] In this study, a high rate of participants (63%) considered adopting different recovery methods depending on the time of the football match (ie, morning, afternoon, and evening). However, no additional questions were provided in our questionnaire. To our knowledge, research on this topic is unknown but warranted.

In contrast to the findings reported by Field et al,[17] we observed that some recovery methods (eg, cold water immersion, active recovery, massage, stretching, and electrostimulation) have been used at later periods after football matches. In the case of cold water immersion, this was only observed after away matches, probably due to logistical issues. However, Brophy-Williams et al[23] and Versey et al[24] reported that cold water immersion provides better benefits in recovery of some performance-related variables when implemented within 30 minutes postexercise, compared with immersions performed some hours after or in the following days. Thus, based on the divergence between the higher

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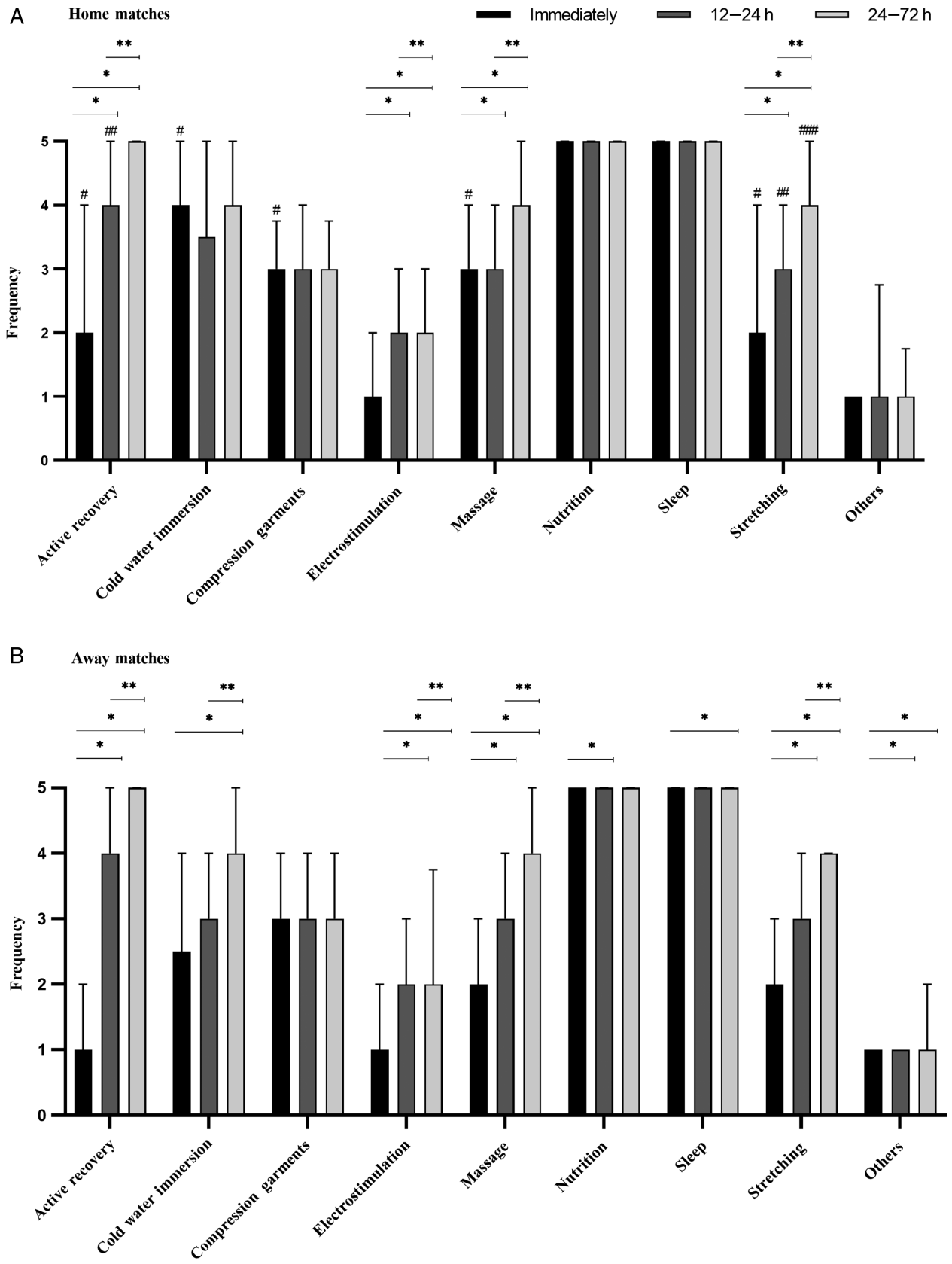

Figure 1 — Degree of recovery methods frequency of use immediately, 12 to 24 hours, and 24 to 72 hours after home (A) and away (B) football matches. Data are shown as median and interquartile range (ie, 75th percentiles).*Significant difference ($P < .017$) from the period immediately after the match. **Significant difference ($P < .017$) from the period 12 to 24 hours after the match. #Significant differences ($P < .05$) between home and away matches, in the period immediately after. ##Significant differences ($P < .05$) between home and away matches, in the period 12 to 24 hours after. ###Significant differences ($P < .05$) between home and away matches, in the period 24 to 72 hours after.

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use of cold water immersion in the later periods of away matches reported in the current study, and the effectiveness shown in literature,[23,24] more research is needed to clarify and to provide recommendations.

In addition, active recovery was used with a greater degree in the periods 12 to 24 hours and 24 to 72 hours after, than immediately after home and away matches. Previous research has concluded that active recovery immediately post football efforts is not effective to recover performance- and perceptive-related parameters,[8,10,25,26] and it may even promote small detrimental effects.[27] Despite the limited evidence, Andersson et al[28] reported similar conclusions when the active recovery was performed later, at 22 and 46 hours after a football match. However, the use of active recovery strategies is normally well accepted, and assumed by athletes to be effective for recovery.[12–15] As the existing studies indicate a low effectiveness of active recovery in the periods after football matches,[8,10,25,26,28] research is needed to determine the reasons underlying the high usage of active recovery in the later periods. With the same trend, massage, stretching, and electroestimulation were used preferably in the later periods, although with low frequency. However, to the best of our knowledge, there are no studies comparing the effect of massage, stretching, and electrostimulation between different postmatch periods. Moreover, according to Abaïdia and Dupont,[8] the level of evidence of these recovery methods is still limited, due to the overall lack of effect described in literature and the few studies reviewed, which may justify the lower use. Specifically, these authors have attributed a grade of recommendation B, which means that the appropriateness of postmatch application of massage, stretching, and electroestimulation is still questionable.

Sleep, nutrition, and cold water immersion (after home matches) showed an overall high and constant degree of usage up to 72 hours after matches. Sleep has been referred to as the most important recovery method performed by teams,[2,7,14,17] which is in line with the current study findings. Strategies like sleeping at least 8 hours after a football match, taking a "power nap" in the days following the match, adjusting training schedules and traveling seem to be critical for a better recovery.[2,7,8,10] However, Nédélec et al[7] reported that sleep deprivation may be worsened in the nights following the match due to the match schedule, associated exposure to bright light, the consumption of stimulating drinks, and travel fatigue. As the frequency of usage of sleep was the highest (ie, 5 points in a Likert scale) and constant among postmatch periods, it remains to be examined whether sleep strategies are similar between postmatch periods, or if special care is needed in each period after the match. This seems to be critical after away matches that might have trips with more than 4-hour duration.

Moreover, nutrition has also been considered a critical recovery method, with a considerable degree of usage.[2,16,17] In the present study, the use of nutrition strategies was high, and constant between postmatch periods in both home and away matches. However, it has been reported that the application of nutrition strategies by teams seems to be more critical immediately postmatch than in the following periods, in particular during congested calendars.[8,11,29] According to Ranchordas et al[11] and Abaïdia and Dupont,[8] the high level of evidence shown in the literature strongly recommends the use of nutritional strategies (ie, grade A; high), in particular immediately after a competitive match, due to the increased activity of glycogen synthesizing enzymes in that period. Otherwise, delaying carbohydrate feeding until 2 hours after exercise can result in lower muscle glycogen concentration by 45% when assessed 4 hours postexercise.[29] Therefore, although frequency of nutrition usage was high and constant among postmatch periods, further research may want to clarify the best nutrition strategies to be taken in the later periods after the match.

Similarly, cold water immersion showed an overall high degree of usage up to 72 hours after home matches. The application of this method has been reported as highly recommended (ie grade A; high),[8] improving perceptive markers of well-being and fatigue.[2,5,30,31] However, the effect of cold strategies in performance and/or physiological outcomes up to 72 hours after exercise-induced muscle damage is still unclear[2,5,30,31] and its long term usage may attenuate the development of muscle mass and strength.[2,32] Due to these unwanted effects, and according to Rey et al,[10] the use of cold water immersion should be preferred in the period immediately after a match (which seems to optimize recovery, than performing at latter postmatch periods)[8,10,25–27] and 2 days before the matchday (to attenuate fatigue effects on the matchday).[10,17] Thus, given the actual scientific recommendation, cold water immersion postmatch practices may need to be adjusted by decreasing its frequency of usage in latter postmatch periods.

Use of Recovery Methods in Different Postmatch Locations

According to previous studies,[7,10,17] match location may also influence the choice of use of recovery methods in team sports. In the present study, we observed that active recovery, cold water immersion, massage, and stretching had lower use in away matches, in particular immediately postmatch. Logistical factors, for instance, may justify the lower use of overall recovery methods immediately after away matches. Despite that most of these methods need minor and transportable equipment, and are easy to use,[10] they require time, which might explain the lower use in away matches. It is, however, important to overcome these limitations, since away matches evoke greater mental and physical fatigue.[7,18] On the other hand, compression garments were reported to be used with slightly higher frequency immediately after away matches. The use of compression garments up to 72 hours after exercise-induced muscle damage has shown a positive effect in muscle swelling, muscle soreness, and lower limb power and strength during recovery.[2,33] According to Abaïdia and Dupont,[8] despite the methodological issues found in the literature, the use of compression garments is recommended (grade A; high) immediately after a match and on the day before. Although with scarce evidence, the use of compression garments combined with electrostimulation was also suggested by Rey et al[10] as an appropriate method immediately after away matches (applied during the trip back home) to optimize recovery. Overall, considering the scant research about the use of recovery methods in home and away matches, future research is needed to determine the reasons why a lower use of recovery methods might exist after away matches compared with home matches.

Study Limitations

Despite the findings and the high participants' adhesion observed, this study has some limitations. First, the exclusive use of a sample of Portuguese football teams does not allow a generalization of the main findings to different contexts worldwide. As such, results may differ depending on the country or culture under study. Second, this research was designed to study the use of recovery methods postmatch in general, and did not consider specific congested

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periods. We are unaware of whether results could differ if questions were directed to this specific context. Finally, questions concerning the recovery methods' frequency of usage did not allow us to obtain information within each recovery method. Consequently, further studies may want to examine the specifications of usage of each method (eg, intensity and duration), which has relevance for understanding periodization of recovery methods.

Practical Applications

- Portuguese professional football (ie, soccer) teams do not use recovery methods following matches in a uniform manner among the recovery periods, supporting the existence of recovery periodization.
- Nutrition and sleep have been recognized by the teams as the most important and frequently used recovery methods postmatch, thus supporting the importance of these recovery methods postmatch.
- A lower usage of recovery methods has been reported after away matches, despite the potential for greater fatigue compared with home matches. This indicates the need to optimize recovery practice after away matches.
- In some recovery methods (eg, active recovery and cold water immersion), the frequency of usage among postmatch periods is not consistent with the current scientific evidence recommendations concerning the most appropriate moments of application. Efforts should be conducted to further explore the inconsistency between theory and practice in professional football.

Conclusions

This study sought to gather new evidence about the use of recovery practices up to 72 hours postmatch in Portuguese football teams. We concluded that the practitioners involved recognized the importance of using recovery methods to mitigate postmatch fatigue, but varying importance has been given to different recovery methods. It was also observed that the recovery methods are not applied uniformly along postmatch periods, and differ depending on the match location (for most of the methods, there are lower use following away matches). Such a fact suggests that recovery methods are periodized following football matches. However, when contrasting the use of methods with the current scientific recommendations about the ideal moment of application to enhance their effects, there is not always consistency. Although more evidence is needed to understand the practices shown in the present study, the practitioners and the athletes should be aware of the need to periodize the recovery methods practices postmatch to improve recovery effectiveness. In addition, the current practices in professional football teams are not enough to provide recommendations, and need to be cross-checked with scientific evidence concerning the recovery methods effectiveness. For instance, the frequency of usage of some recovery methods (eg, active recovery and cold water immersion) in postmatch periods may not be consistent with the current scientific evidence concerning the most appropriate moments of application.

Acknowledgments

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors. The authors thank all soccer clubs and professionals for participating in this study. The original contributions presented in the study are included in the article/Supplementary Data (Files 1 and 2) (available online), further inquiries can be directed to the corresponding author/s. The studies involving human participants were reviewed and approved by the ethics committee of the Faculty of Human Kinetics, University of Lisbon (approval number: 10/2019). All participants provided written informed consent to participate in this study, and the procedures were conducted according to the principles expressed in the Declaration of Helsinki.

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